

Tutorial Questions (Set-VII)

- Q1 (i) Let $\varphi_n: \mathbb{N} \rightarrow \mathbb{R}$ be given by $\varphi_n(m) = \frac{m}{n+m}$ for $m, n \in \mathbb{N}$. Show that the pointwise limit of the sequence $(\varphi_n)_{n \in \mathbb{N}}$ is the zero function. Conclude that

$$\lim_{m \rightarrow \infty} \left(\lim_{n \rightarrow \infty} \varphi_n(m) \right) \neq \lim_{n \rightarrow \infty} \left(\lim_{m \rightarrow \infty} \varphi_n(m) \right).$$

- (ii) Let (E, d) be a metric space and $f_n: E \rightarrow \mathbb{R}$ for $n \in \mathbb{N}$. Suppose $f_n \rightarrow f$ uniformly on E . If x is an accumulation point of E and $\lim_{t \rightarrow x} f_n(t) = A_n$ for $n \in \mathbb{N}$, then show that the sequence $(A_n)_{n \in \mathbb{N}}$ is convergent and $\lim_{t \rightarrow x} f(t) = \lim_{n \rightarrow \infty} A_n$. In other words, interchange of limits is valid for uniform convergence.

- (iii) Let $g_n(x) = \frac{x^{2n}}{x^{2n}+1}$ for $x \in \mathbb{R}$ and $n \in \mathbb{N}$. Let $g(x) = \begin{cases} 0 & \text{if } |x| < 1, \\ \frac{1}{2} & \text{if } |x| = 1, \\ 1 & \text{if } |x| > 1. \end{cases}$ Show

that $g_n \rightarrow g$ pointwise on $\mathbb{R}$ but convergence is not uniform.

- (iv) Let $f_n(x) = nx(1-x^2)^n$ for $x \in [0, 1]$. Show that the pointwise limit of $(f_n)_{n \in \mathbb{N}}$ is the zero function, but convergence is not uniform on $[0, 1]$. Verify that $\lim_{n \rightarrow \infty} \int_0^1 f_n dx \neq \int_0^1 \left(\lim_{n \rightarrow \infty} f_n \right) dx$.

- (v) Let $h_n(x) = \frac{\sin nx}{\sqrt{n}}$ for $x \in \mathbb{R}$ and $n \in \mathbb{N}$. Show that $h(x) = \lim_{n \rightarrow \infty} h_n(x) = 0$. Verify that the derivative sequence $h'_n \not\rightarrow h'$. Here $h_n \rightarrow h$ uniformly on $\mathbb{R}$.

- (vi) Let $f_n(x) = \frac{x^2}{x^2+(1-nx)^2}$ for $x \in [0, 1]$ and $n \in \mathbb{N}$. Show that $f_n \rightarrow 0$, the zero function in pointwise manner, but convergence is not uniform.

- Q2 (i) Let K be a compact metric space and $\mathcal{F} = \{F_\alpha : \alpha \in \Lambda\}$ be a family of closed subsets of K . Suppose for every finite subset $\{\alpha_1, \alpha_2, \dots, \alpha_n\} \subseteq \Lambda$, the intersection $F_{\alpha_1} \cap F_{\alpha_2} \cap \dots \cap F_{\alpha_n} \neq \emptyset$, i.e., $\mathcal{F}$ has the *Finite Intersection Property*. Then show that the intersection $\bigcap_{\alpha \in \Lambda} F_\alpha \neq \emptyset$. Furthermore, the converse also holds.

- (ii) (Dini's Theorem) Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of continuous real-valued function on a compact metric space K . Suppose $f_n \rightarrow f$ converges pointwise on K and $f_n(x) \geq f_{n+1}(x)$, $\forall x \in K$ and $n \in \mathbb{N}$. Show that $f_n \rightarrow f$ converges uniformly on K .

- (iii) Consider the sequence $(f_n)_{n \in \mathbb{N}}$ on $K = (0, 1)$ given by $f_n(x) = \frac{1}{nx+1}$. The sequence $(f_n)_{n \in \mathbb{N}}$ satisfies all conditions of the Dini's theorem except the compactness of $K = (0, 1)$. Show that in this case, convergence is not uniform.

Q3 (i) (Weierstrass M-test) Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of real-valued function on E . Suppose $|f_n(x)| \leq M_n$, $\forall x \in E$ and $n \in \mathbb{N}$. If $\sum_{n=1}^{\infty} M_n < \infty$, then show that the series $\sum_{n=1}^{\infty} f_n$ is uniformly convergent on E . Construct an example of a sequence $(f_n)_{n \in \mathbb{N}}$ such that the series $\sum_{n=1}^{\infty} f_n$ is uniformly convergent but $\sum_{n=1}^{\infty} M_n$ is not convergent.

(ii) Suppose $f_n \in \mathcal{R}[a, b]$ for $n \in \mathbb{N}$ and the series $\sum_{n=1}^{\infty} f_n$ converges uniformly to a function f on $[a, b]$. Then show that $f \in \mathcal{R}[a, b]$ and $\int_a^b f \, dx = \sum_{n=1}^{\infty} \int_a^b f_n \, dx$.

(iii) Suppose $(f_n)_{n \in \mathbb{N}}$ is a sequence of continuously differentiable functions on $[a, b]$. Suppose the derivative sequence $(f'_n)_{n \in \mathbb{N}}$ converges uniformly to a function g on $[a, b]$ and the sequence $(f_n(x_0))_{n \in \mathbb{N}}$ is convergent for some point $x_0 \in [a, b]$. Then show that $(f_n)_{n \in \mathbb{N}}$ converges uniformly to a function f on $[a, b]$ and

$$f'(x) = g(x) = \lim_{n \rightarrow \infty} f'_n(x), \quad \forall x \in [a, b].$$

Q4 (i) Show that the sequence $\{f_n : n \in \mathbb{N}\}$ of functions on $[0, 1]$ is NOT an equicontinuous family, if a) $f_n(x) = x^n$, b) $f_n(x) = \frac{x^2}{x^2 + (1-nx)^2}$, c) $f_n(x) = \begin{cases} nx, & 0 \leq x \leq \frac{1}{n}, \\ 1, & \frac{1}{n} \leq x \leq 1. \end{cases}$

(ii) Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a continuous function. Suppose $f_n(x) = f(nx)$ for $x \in [0, 1]$ and $\{f_n : n \in \mathbb{N}\}$ is an equicontinuous family. Show that f is constant.

Q5 (Optional) Let (X, d) be a metric space and $B(X)$ be the set of bounded real-valued functions on X .

- (i) Show that $B(X)$ is a vector space over $\mathbb{R}$ under pointwise operations.
- (ii) Define $\|f\|_{\infty} = \sup\{|f(x)| : x \in X\}$ for $f \in B(X)$. Show that $\|\cdot\|_{\infty}$ is a norm function on $B(X)$ called the *sup norm*. Deduce that $B(X)$ is a metric space under the sup-metric $d_{\infty}(f, g) = \|f - g\|_{\infty}$ for $f, g \in B(X)$, induced by the sup norm $\|\cdot\|_{\infty}$.
- (iii) Show that a sequence $(f_n)_{n \in \mathbb{N}}$ in $B(X)$ converges to f in the metric space $B(X)$ under the sup-metric if and only if $f_n \rightarrow f$ converges uniformly on X .
- (iv) Show that $B(X)$ is a complete metric space.
- (v) Let $C_b(X)$ be the set of continuous, bounded real-valued functions on X . Show that $C_b(X)$ is a vector subspace of $B(X)$. Also, prove that $C_b(X)$ is closed in $B(X)$. Deduce that $C_b(X)$ is also a complete metric space under sup-metric.